

Full answers to two multilinear type questions

Candidate full result for arXiv:1502.00440

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Status. Candidate full result, likely valid, requiring human review. Both open questions in Botelho-Campos are answered affirmatively. The first answer is stronger because the sign lower estimate is unnecessary; the second is stronger because the noninclusion holds even if all factorization exponents equal one.

1 Definitions and the two questions

An m -linear map $T : E_1 \times \cdots \times E_m \rightarrow F$ has type $(p_1, \dots, p_m)$ if a constant C satisfies

$$\left(\int_0^1 \left\| \sum_{j=1}^k r_j(t) T(x_j^{(1)}, \dots, x_j^{(m)}) \right\|^2 dt \right)^{1/2} \leq C \prod_{i=1}^m \left(\sum_{j=1}^k \|x_j^{(i)}\|^{p_i} \right)^{1/p_i}. \quad (1)$$

The tuple is *proper* when

$$\frac{1}{2} \leq \sum_{i=1}^m \frac{1}{p_i} < 1. \quad (2)$$

Write τ_p for the classical ideal of linear type- p operators. The factorization multi-ideal $\mathcal{L}(\tau_{r_1}, \dots, \tau_{r_n})$ consists of maps $B(u_1, \dots, u_n)$ with u_i of linear type r_i .

The source asks first about the tensor map

$$T = \varphi_1 \otimes \cdots \otimes \varphi_n \otimes id_E, \quad T(x_1, \dots, x_{n+1}) = \varphi_1(x_1) \cdots \varphi_n(x_n) x_{n+1},$$

where E has type p and the sign lower p -estimate property.

In part (a) of the proposition above, for $1 < p < 2$ it is always possible to choose $r_1, \dots, r_{n+1}$ with $\frac{1}{2} \leq \frac{1}{r_1} + \cdots + \frac{1}{r_{n+1}} < \frac{1}{p}$, and then $\varphi_1 \otimes \cdots \otimes \varphi_n \otimes id_{E_{n+1}}$ fails to have some proper type $(r_1, \dots, r_{n+1})$.

Open question 1. Under the assumptions of Proposition [2.12](#)(a), does $\varphi_1 \otimes \cdots \otimes \varphi_n \otimes id_{E_{n+1}}$ have some/any proper type $(r_1, \dots, r_{n+1})$ with $\frac{1}{p} \leq \frac{1}{r_1} + \cdots + \frac{1}{r_{n+1}}$?

Figure 1: Open Question 1 in the source PDF [1, p. 9].

It then asks whether every factorization ideal with $1 \leq r_i \leq 2$ and at least one $r_i > 1$ fails to be contained in every proper multilinear type ideal.

Open question 2. Is it true that $\mathcal{L}(\tau_{r_1}, \dots, \tau_{r_n}) \not\subseteq \tau_{p_1, \dots, p_n}$ for every proper type $(p_1, \dots, p_n)$ for n -linear operators and all $1 \leq r_1, \dots, r_n \leq 2$ with $r_j > 1$ for some j ?

Figure 2: Open Question 2 in the source PDF [1, p. 12].

2 Every tuple above the type threshold

Theorem 1 (Answer to Open Question 1). *Let $1 < p \leq 2$, let E have type p , and let $\varphi_i \in E'_i$ for $1 \leq i \leq n$. If $(r_1, \dots, r_{n+1})$ is any proper type satisfying*

$$\sum_{i=1}^{n+1} \frac{1}{r_i} \geq \frac{1}{p}, \quad (3)$$

then $\varphi_1 \otimes \dots \otimes \varphi_n \otimes id_E$ has type $(r_1, \dots, r_{n+1})$. Quantitatively,

$$\|T\|_{\tau_{r_1, \dots, r_{n+1}}} \leq T_p(E) \prod_{i=1}^n \|\varphi_i\|, \quad (4)$$

where $T_p(E)$ is a type- p constant of E . In particular, the sign lower p -estimate assumption is not needed.

Proof. Put

$$S = \sum_{i=1}^{n+1} \frac{1}{r_i}, \quad s = S^{-1}.$$

Condition (3) says $s \leq p$. Also $s \leq r_i$ for every i , and $\sum_i s/r_i = 1$.

For finite families $x_j^{(i)} \in E_i$ ($i \leq n$) and $x_j^{(n+1)} \in E$, type p of E gives

$$\begin{aligned} & \left\| \left(T(x_j^{(1)}, \dots, x_j^{(n+1)}) \right)_{j=1}^k \right\|_{\text{Rad}(E)} \\ & \leq T_p(E) \left(\sum_{j=1}^k \left[\prod_{i=1}^n |\varphi_i(x_j^{(i)})| \|x_j^{(n+1)}\| \right]^p \right)^{1/p}. \end{aligned}$$

Since $s \leq p$, the ℓ_s norm dominates the ℓ_p norm. Generalized Hölder with exponents r_i/s now yields

$$\begin{aligned} & \left(\sum_{j=1}^k \left[\prod_{i=1}^n |\varphi_i(x_j^{(i)})| \|x_j^{(n+1)}\| \right]^s \right)^{1/s} \\ & \leq \prod_{i=1}^n \left(\sum_{j=1}^k |\varphi_i(x_j^{(i)})|^{r_i} \right)^{1/r_i} \left(\sum_{j=1}^k \|x_j^{(n+1)}\|^{r_{n+1}} \right)^{1/r_{n+1}} \\ & \leq \left(\prod_{i=1}^n \|\varphi_i\| \right) \prod_{i=1}^{n+1} \left\| (x_j^{(i)})_{j=1}^k \right\|_{\ell_{r_i}(E_i)}, \end{aligned}$$

where $E_{n+1} = E$. Combining the two displays proves (1) and (4). $\square$

Remark 2 (The boundary is included). *At equality in (3), one uses type p itself. If only the supremal type index of E were p and the endpoint were not attained, the same proof would cover the strict inequality but not automatically the boundary. The source hypothesis explicitly says that E has type p .*

3 One factorized map outside every proper type ideal

Theorem 3 (Answer to Open Question 2). *Let $n \geq 2$ and $1 \leq r_i \leq 2$ for $1 \leq i \leq n$. Define*

$$B : \ell_{r_1} \times \cdots \times \ell_{r_n} \longrightarrow \ell_1, \quad B(x^{(1)}, \dots, x^{(n)})_m = \prod_{i=1}^n x_m^{(i)}. \quad (5)$$

Then $B \in \mathcal{L}(\tau_{r_1}, \dots, \tau_{r_n})$, but B has no proper multilinear type. Consequently, for every proper type $(p_1, \dots, p_n)$,

$$\mathcal{L}(\tau_{r_1}, \dots, \tau_{r_n}) \not\subseteq \tau_{p_1, \dots, p_n}.$$

This remains true when $r_1 = \cdots = r_n = 1$.

Proof. Since $r_i \leq 2$ and $n \geq 2$, $\sum_i 1/r_i \geq n/2 \geq 1$. Choose numbers $b_i \in [0, 1/r_i]$ with $\sum_i b_i = 1$, and put $q_i = 1/b_i$, interpreting $q_i = \infty$ if $b_i = 0$. Then $q_i \geq r_i$. Generalized Hölder and the contractive inclusion $\ell_{r_i} \hookrightarrow \ell_{q_i}$ give

$$\|B(x^{(1)}, \dots, x^{(n)})\|_1 \leq \prod_{i=1}^n \|x^{(i)}\|_{q_i} \leq \prod_{i=1}^n \|x^{(i)}\|_{r_i}. \quad (6)$$

Thus B is continuous.

For $1 \leq r \leq 2$, the identity of ℓ_r has linear type r (at $r = 1$ this is the triangle inequality; for $1 < r \leq 2$ it is the classical type of ℓ_r). The factorization

$$B = B(id_{\ell_{r_1}}, \dots, id_{\ell_{r_n}})$$

therefore places B in $\mathcal{L}(\tau_{r_1}, \dots, \tau_{r_n})$.

Now take $x_j^{(i)} = e_j$ for $1 \leq j \leq k$ and every i . Then $B(e_j, \dots, e_j) = e_j$, and

$$\|(e_j)_{j=1}^k\|_{\text{Rad}(\ell_1)} = \left(\int_0^1 \left\| \sum_{j=1}^k r_j(t) e_j \right\|_1^2 dt \right)^{1/2} = k. \quad (7)$$

If B had type $(p_1, \dots, p_n)$, (1) would imply

$$k \leq C \prod_{i=1}^n k^{1/p_i} = C k^{\sum_i 1/p_i}.$$

For a proper type the exponent is strictly below one by (2), a contradiction as $k \rightarrow \infty$. $\square$

4 Scope, verification, and novelty boundary

Question 1 is solved for *every* qualifying tuple, including equality at the threshold, and under the weaker assumption that E merely has type p . Question 2 is solved by one explicit operator for each factorization tuple; that operator simultaneously avoids the union of all proper diagonal type ideals. No assertion about the paper's multilinear cotype ideals is made.

The proof has no computational component. Its review-critical equalities are the monotone embedding $\ell_s \hookrightarrow \ell_p$ in Theorem 1, the choice of q_i in (6), and the exact Rademacher norm (7). All endpoints $p = 2$, $r_i = 1$, and $r_i = 2$ are included.

The current source is arXiv:1502.00440v2, last revised 21 December 2015, and still contains both questions. The run indexes and bounded searches through 11 August 2026 for the exact question labels, title-plus-question phrases, factorization notation, and coordinatewise-product counterexamples found no later answer. Both proofs are elementary consequences of the definitions and could be folklore or overlooked; no priority claim is made.

Human-review recommendation. Check the direction of the sequence-space embedding in Theorem 1 and verify that $b_i \in [0, 1/r_i]$ with sum one always exist. Then reproduce the diagonal basis calculation (7). Subject to those checks, classify both answers as full and affirmative.

References

- [1] G. Botelho and J. R. Campos, *Type and cotype of multilinear operators*, arXiv:1502.00440v2 (2015).
- [2] J. Diestel, H. Jarchow, and A. Tonge, *Absolutely Summing Operators*, Cambridge University Press, 1995; see the classical type of ℓ_r .